🌻 Individual questions – introduction

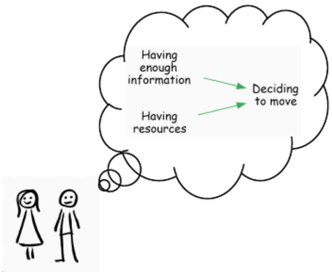
Chapter contents.

Because the output is a structured network, we can apply a range of queries to explore the data. This gives us a library of **pre-existing approaches** to ask **practical questions** about the causal landscape described by the participants.

Here's a poster showing a few of the main questions.

Evaluation questions you can answer with causal mapping

Maybe you've heard of causal mapping...



But can you answer useful evaluation questions with it?

Here are a few examples! 👉

What is causal mapping?

Non-causal Qualitative Data Analysis (QDA) version 1: code causality as a theme/tag/concept

Text and quote	Theme / tag
"the floods destroyed our crops"	Floods destoy crops (causal)

Non-causal QDA version 2: code themes/tags/concepts, with optional post-hoc connections between themes

Text and quote	Theme / tag
"the floods destroyed our crops"	Floods
"the floods destroyed our crops"	Crops destroyed

Optional post-hoc connection

Causal QDA: code causal claims natively between *pairs* of codes ✅

Text and quote	"Cause"	"Effect"
"the floods destroyed our crops"	Floods happened	Crops destroyed

Quote	Cause	Effect	Source ID
The floods destroyed our crops	Floods happened	Crops destroyed	A3
the chickens died too	Floods happened	Chickens died	A3
Our harvest was ruined due to the high water	Floods happened	Crops destroyed	A4

We code many "causal claims" across many sources.

We use a codebook for cause/effect labels ("factors") which is either predefined or which we develop iteratively

The result is a table of causal links which can be visualised as a **map**

Links and factors can be labelled e.g. with ***citation count****, meaning how many times it was mentioned in the map as currently filtered, or* ***source count*** *(how many different sources mentioned it).*
We can also use colours and sizes to reflect different metrics like these.

What is a causal map?

A **causal map** is a diagram in which boxes (which we call factors) are joined by directed links. A link means that someone claimed that one thing influenced or contributed to another.

A **systems map** usually tries to model *reality*. A **causal map** (or "causal concept map") usually tries to model *how stakeholders think about reality*. This approach has been used for 50 years (Axelrod, Eden, Ackerman…) in many disciplines, though not so much in evaluation.

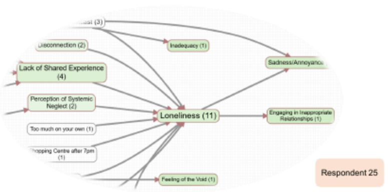
DO use causal mapping when you…

- have many causal claims about links between many causal factors from many different sources
- want to organise all these causal claims
- are interested in differences between sources

DON'T use causal mapping when you…

- are interested in a small number of specific high-stakes causal connections between a small number of causal factors, like the link between "taking the medicine" and "feeling better" (try Process Tracing instead)
- want a mathematical model, e.g. for predicting future states of the system (Try Causal Loop Diagramming instead)
- would like to collaboratively develop a model of a problem or a plan for an intervention without worrying too much about who said what (try PRSM instead)

View individual pathways:
What is one respondent's experience?



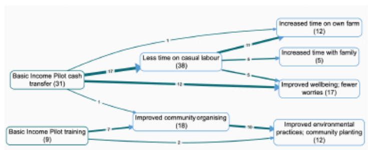
Identify individual perspectives and narratives on project mechanisms. Get a picture of their worldview.

Understand individual viewpoints to address specific concerns or leverage unique knowledge.

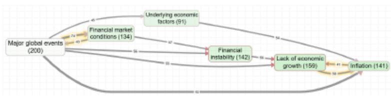
Show all the individual views together as an aggregated map or filter just to show one source or group. 👉

View aggregated pathways:
What is the main story?

Causal mapping can help you identify and visualise common features of the causal stories told by most stakeholders. The birds-eye view. It's a process which is creative (establishing your codebook) yet data-driven.



Can we find positive **feedback loops** within the network?



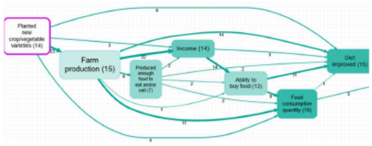
Compare groups:
Which links were mentioned more by some groups than others?

Query a causal map to **compare link frequencies between groups** and find links with surprising differences between groups, controlling for baseline frequencies. Above we see that these men mention a link between increased knowledge and health behaviour more frequently than these women.

Factor	*Female	*Male	Significant (All)	Citation Count
Increased knowledge	147	83	No	230
Farm production	107	52	No	159
Health behaviour	81	70	Yes	151
Improved health	87	58	No	145
Diet improved	67	34	No	101

The frequency with which a **link between factors** is mentioned can, **with caveats**, be interpreted as a strength of the evidence for that link, contributing towards a final evaluative judgement against your rubrics.

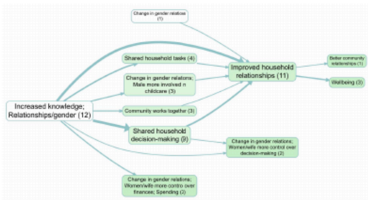
Trace causal chains:
What are the direct and indirect consequences of one or more factors?



Trace **chains of influence** from a starting point like an intervention to a key outcome, or from an outcome back to drivers, revealing the step-by-step or branching logic described by the sources.

Beware the *transitivity trap!* Be careful when joining individual pathways into longer causal claims if they come from different sources in different contexts. You can use "source tracing" to mitigate this but can be quite tricky to do. You need patience, or specialised software.

Identify unexpected elements:
What are the emerging factors?



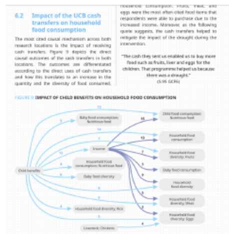
If you code your transcripts or documents without reference to any 'official' codebook or theory of change, you can discover how your sources conceive of the key causal factors and the words they use to describe them.

It's useful for **identifying unexpected or evolving elements** in a project's ecosystem and to understand how different factors combine and interact in ways you might not have anticipated.

Impact pathways:
What is the evidence that the intervention influenced key outcomes?

Causal mapping doesn't give you 'The Answer,' but is a way of **organising evidence to help you make final evaluation judgements and causal inferences.**

- Compare causal maps with your theorfy of change
- Compare the strength of evidence for different pathways and what this means for your rubrics
- Use original quotes to understand people's experiences



Causal mapping helps to organise the evidence. The judgement is still yours.

Find out more about our passion for causal mapping

Causal Map 4, our app for causal mapping Core functionality is free for unlimited public projects!

app.causalmap.app

Steve Powell

Fiona Remnant

Come and talk to us about causal mapping!

Our AI interviewer, QualiaInterviews

QualiaInterviews.com

Full guide to causal mapping

garden.causalmap.app

See also

- Formatting your map for what you want to show for the visualisation side.
- Vignettes for textual outputs.

Focusing on specific factors. What influences and outcomes are connected to a specific factor ?

Looking downstream. What are the direct and indirect consequences of one or more factors ?

Looking upstream. What are the direct and indirect influences on one or more factors ?

Path tracing – How do one or more causes affect one or more effects, including indirect pathways ?

Source tracing – What are the consequences of one or more factors, looking only at stories told in their entirety by individual sources ?

Robustness – How robust is the evidence for that X influences Y ?

Counting and comparing influences

Properties of the causal map – Which factors are reported as being causally central or causally peripheral ?

Properties of the causal map – What is the overall structure of the network ?

Properties of the causal map – Are there leverage points ?

Properties of the causal map – Are there feedback loops ?

Combining questions

Tribes. The most relevantly different subgroups in your data (by causal story)

Showing group data as custom link labels on the map with optional significance test

Names of tables and fields

Hierarchical coding